Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Inconsistent convergence in Hilbert space

Files.

- File: `briefs/InconsistentConvergence/InconsistentConvergence.lean`
- Library: `InconsistentConvergence`
- Imports: `Mathlib.Analysis.Normed.Lp.lpSpace`

Setting. $H = \ell^2(\mathbb{N})$ (`lp (fun _ : ℕ => ℂ) 2`), with $|n\rangle = \delta_n$ the oscillator eigenstates and

$$|e^j\rangle = \sqrt{\tfrac{j-1}{j}}\;|0\rangle + \sqrt{\tfrac{1}{j}}\;|j\rangle\,.$$

Results.

Brief	Lean
eq. (2): $\|e^j - \|0\rangle\|^2 = \left(\sqrt{(j-1)/j} - 1\right)^2 + 1/j$	`norm_sub_ground_sq`
$e^j \to \|0\rangle$ in ℓ^2	`tendsto_e`
eq. (3): $\sum_n n^2 \|e^j_n\|^2 = j \to \infty$	`secondMoment_e`, `tendsto_secondMoment_e`
the $\|j\rangle$ alone have no limit in ℓ^2	`not_cauchySeq_level`: $\| \|j\rangle - \|k\rangle \|^2 = 2$ for $j \neq k$
eq. (4): $\langle e^j \| N \| e^j \rangle = 1$ for $j \geq 1$	`numberExp_e`
eq. (5), the inconsistency	`not_continuousAt_numberExp`: $\langle N \rangle$ is not continuous at $\|0\rangle$ in ℓ^2

Differences from the brief.

- Indexing. Proposition 1 indexes sequences from 1, with the ground term at $i = 1$. The formalization indexes from 0, matching the oscillator of Section 3. The divergence is the same; with the brief's indexing the sum is $(j-1)/j + j$ instead of j.
- No limit in $S(\mathbb{N})$. What is proved is that the seminorm $\sum_n n^2 |x_n|^2$ is unbounded along e^j, which is the brief's argument. The Schwartz sequence space $S(\mathbb{N})$ and its topology are not constructed.
- $\langle N \rangle$ is defined on all of ℓ^2 as the weighted sum $\sum_n n\,|x_n|^2$, which has a default value outside the domain of N. The discontinuity is still meaningful, because every e^j is finitely supported.

Problems found in the brief. No mathematical problems. Minor: the distance is written $d(x, y) = \sum_i (x_i - y_i)^2$, which is the squared distance and, for complex entries, should read $\sum_i |x_i - y_i|^2$.